

# AROOJ FATIMA

AI Engineer | NLP & Generative AI | MLOps

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## PROFESSIONAL SUMMARY

Results-driven AI Engineer with of hands-on experience designing and deploying intelligent AI systems at production scale. Proven expertise in RAG pipelines, LLM fine-tuning, multi-agent architectures, and NLP. Delivered end-to-end solutions — from government AI programs to voice-enabled LMS assistants — combining research-driven methodology with practical MLOps deployment on AWS and Google Cloud.

## CORE SKILLS

|            |                                                                            |
|------------|----------------------------------------------------------------------------|
| Languages  | Python, SQL                                                                |
| AI / ML    | NLP, Generative AI, LLMs (Gemini, OpenAI, Llama, Falcon), Machine Learning |
| Frameworks | TensorFlow, LangChain, LangGraph, CrewAI, Hugging Face, Google ADK         |
| Tools      | Git, Docker, Jira, Streamlit, Flask                                        |
| Deployment | MLOps, AWS (EC2, ECR), Google Cloud, CI/CD (GitHub Actions)                |

## PROFESSIONAL EXPERIENCE

AI Engineer | [Corvit Networks](#) Nov 2025 — Present

- Architected an intelligent career guidance bot for the **Digital Youth Hub**, integrating multi-platform data to deliver personalised career roadmaps and job matches; supplemented by an admin-side analytics engine surfacing sentiment trends and quality improvement reports.
- Engineered a **voice-enabled AI assistant** for a campus LMS, supporting natural-language commands across system modules (attendance, grading) with real-time insights on pending assignments and upcoming events.
- Built an end-to-end Sales Automation System for a networking firm, automating the full cycle from email query intake through SQR, quotation, proposal, and tender generation — cutting manual workforce requirements by **80%**.

Data Scientist | [DataWorld](#) | [Project Collaboration Program](#) Jan 2025 — Apr 2025

- Developed a Credit Card Time-Series forecasting model using Facebook Prophet, improving predictive accuracy on transactional data.
- Conducted FIFA dataset EDA on large-scale database records, engineering a match result predictor and defensive pressure classifier.
- Identified key cardiovascular risk indicators (cholesterol, BP, age) through statistical analysis of heart disease datasets.
- Designed and evaluated a Credit Card Fraud Detection pipeline, applying anomaly detection techniques to flag high-risk transactions.

## TEACHING EXPERIENCE

Junior Lecturer | [University of Management & Technology \(UMT\)](#) Sep 2025 — Feb 2026

- Delivered lectures and lab sessions; developed teaching resources that improved student comprehension of core CS concepts.

Teaching Assistant — NLP, OS & DBMS | [UMT](#) Oct 2024 — Jul 2025

- Mentored students in NLP (NLTK, TensorFlow), OS process scheduling, and database design; reduced project revision cycles by providing structured code reviews and model debugging support.

## KEY PROJECTS

LegalAI.Pak *Final Year Project — UMT*

- Designed a domain-specific legal RAG system for Pakistani law with automated ingestion (PDF/DOCX/TXT), FAISS + BM25 hybrid indexing, and reranking — achieving **85% Top-1 and 99% on Top-5 retrieval accuracy** and **MRR of 0.92**.
- Implemented citation verification (court match, year match, recall/precision) and an argument builder, validated by RAGAS-style metrics to ensure hallucination-free, trustworthy outputs for legal professionals.
- Engineered a full-stack legal platform with lawyer profile management, a case workspace (real-time communication, milestone tracking, secure document sharing), and a community Q&A forum.

Tech: Python, OpenAI API, LangChain, FAISS, Pinecone, Flask, ReactJS, ExpressJS, SQL

### SmartShop.AI [Capstone](#) — [Kaggle GenAI Course](#)

- Built a multi-agent recommendation system with a 3-layer RAG pipeline (scraping → vector embedding → AI summarisation) to deliver sentiment-enriched product analysis, deployed with a Streamlit frontend and low-latency voice interface.

Tech: CrewAI, Gemini, Whisper, Render

### Sonic Guide.AI [Hackathon](#) — [The Internet of Agents](#)

- Developed a multi-agent platform generating personalised, real-time audio tours for global locations by orchestrating specialised agents for architecture, history, culinary, and culture content.

Tech: Python, OpenAI Agents, GPT-4o, TTS, Streamlit

### Medical Chatbot [Hackathon](#) — [GenAI Project](#)

- Deployed a production-grade medical chatbot on AWS (EC2/ECR) with Docker and GitHub Actions CI/CD, leveraging a curated medical knowledge base for preliminary diagnosis and responsible AI safeguards.

Tech: GPT-3.5, Pinecone, LangChain, Flask, AWS EC2/ECR, Docker, GitHub Actions

## EDUCATION

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### Bachelor of Software Engineering (BSSE)

2021 — 2025

University of Management & Technology | **CGPA: 3.62 / 4.0**

## CERTIFICATIONS

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Generative AI Engineer — [Kaggle](#)

Google Data Scientist — [Udemy](#)

Certified Python Developer — [PFTP](#)

LSTWSTC'25 Speaker — [IEEE](#)

## AWARDS & SCHOLARSHIPS

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- Merit-Based Scholarship — **90% tuition fee waiver** awarded by UMT for outstanding academic performance.
- Dean's Award — Recognised for achieving **3.93 SGPA** in Fall 2023.